

**ACTIVITY 13 — LEARNER WORKSHEET****Solution Selection Matrix, Benchmarking, and FMEA**

| TYPE              | DURATION | DMAIC PHASE |
|-------------------|----------|-------------|
| Elective Activity | 60 min   | IMPROVE     |

|      |  |
|------|--|
| Name |  |
| Date |  |

**What you are doing**

Three Improve-phase tools from the original course. Score solutions against weighted criteria, benchmark against outstanding practice, and use FMEA to find how a proposed change could fail before it does.

**What you will produce:** A scored solution selection matrix ranked by weighted score, a benchmarking summary and an FMEA with RPN and actions on the highest scores.

**Tools and techniques:** Solution selection matrix, benchmarking, FMEA, RPN scoring

**The scenario**

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

**Files in this activity folder****data/ — the mock data you analyse:**

- candidate-solutions.csv — 5 rows (solution\_id, candidate\_solution, est\_cost, est\_effort\_days)

**templates/ — the worksheets you complete:**

- fmea.csv (process\_step, failure\_mode, effect, cause, severity\_1\_10, occurrence\_1\_10, detection\_1\_10, RPN, action)
- solution-selection-matrix.csv (solution, impact\_w5, cost\_w3, ease\_w2, weighted\_score, rank)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

### Your steps

1. Agree the selection criteria and weight each by importance.
2. Score every candidate solution against the weighted criteria and rank the results.
3. Benchmark your process against an internal or external reference and note the gap.
4. Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detection.
5. Calculate  $RPN = S \times O \times D$  and address the highest-RPN failure modes first.

### Your working space

Complete every field. Keep this worksheet — it is your reference for applying the method at work.

#### Step 1 — Agree the selection criteria and weight each by importance.

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#### Step 2 — Score every candidate solution against the weighted criteria and rank the results.

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#### Step 3 — Benchmark your process against an internal or external reference and note the gap.

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#### Step 4 — Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detec...

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#### Step 5 — Calculate $RPN = S \times O \times D$ and address the highest-RPN failure modes first.

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## Answers to the discussion prompts

1. Did the weighted score change the ranking versus your gut preference?

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2. Which failure mode has the highest RPN, and is it the one you feared?

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3. Is a high-severity, low-occurrence risk more dangerous than the reverse?

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## Reflect & discuss

What change at your workplace failed in a way an FMEA would have caught?

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Who would you benchmark against, and would they talk to you?

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## Check your work

Your matrix ranks solutions by weighted score, and every FMEA row has an RPN and an action for the highest scores.